

Gaurav Girish Rathod

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Portfolio | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

- IIT Mandi undergraduate specializing in applied machine learning, computer vision, and self-supervised learning systems. Proven experience designing multimodal neural architectures, gradient-boosted boundary estimation pipelines, and optimization algorithms with PyTorch and Python. Experienced in deploying models across large datasets (180k+ samples), optimizing cross-domain generalization, and benchmarking performance.

EDUCATION

- **Indian Institute of Technology, Mandi** Himachal Pradesh, India
B.Tech in Civil Engineering with Second Major in Computer Science; CGPA: 8.34 *August 2024 – Present*
 - **Core Coursework:** Machine Learning, Deep Learning & Applications, Computing & Data Science, Linear Algebra, Probability & Statistics, Design & Analysis of Algorithms, Data Structures, Operating Systems
- **Queen Mary Junior College** Mumbai, Maharashtra, India
Higher Secondary Certificate (HSC); Percentage: 85.5% *2022 – 2024*

TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** PyTorch, Scikit-Learn, Hugging Face, Torchvision, Optimal Transport (POT), LLMs, RAG, SSL
- **Computer Vision & OCR:** OpenCV, Vision Transformers (ViT), Feature Fusion, PaddleOCR, Anomaly Detection, Image Augmentation
- **Languages & Core:** Python, C++17, C, SQL (PostgreSQL, MySQL), Bash, Data Structures & Algorithms
- **Data, MLOps & Tooling:** Pandas, NumPy, Playwright, Git, Docker, Linux/Unix, RESTful APIs, CI/CD, GTest, GDB

TECHNICAL PROJECTS

- **Robust Deepfake Detection Under Domain Shift** [GitHub](#) | [Live](#)
PyTorch, DINOv2, Optimal Transport (FGW), SRM Filters, Python *2026*
 - **Multimodal Feature Fusion:** Engineered a dual-branch neural architecture coupling a frozen 307M DINOv2 ViT-L/14 with a 30-kernel SRM noise stream, fusing 1024-d semantic tokens and 128-d frequency descriptors into an 1153-d classification head.
 - **Graph Matching:** Constructed a 32-prototype representative face-graph memory bank and executed Fused Gromov-Wasserstein (FGW) matching via a 20-iteration Sinkhorn solver to yield 224×224 spatial anomaly activation maps.
 - **Cross-Domain Generalization:** Attained **0.884 cross-domain AUC** (vs. 0.831 baseline, +5.3%) and **0.946 validation AUC** (88.7% accuracy), mitigating severe accuracy drop-offs under multi-codec compression across 140k+ test samples.
- **Process Discovery & HITL ML Automation Engine** [GitHub](#)
Python, Scikit-Learn, PaddleOCR, Process Mining, Playwright *2026*
 - **Temporal Boundary Detection:** Developed an automated event boundary model on 183k+ operation sequences using HistGradientBoosting, securing a 0.905 F1-score and aggregating 540 unlabeled work units with 1.10s mean timestamp deviation.
 - **OCR & Clustering:** Formulated a multimodal feature pipeline leveraging PaddleOCR (> 0.97 confidence) across 1.1k+ UI frames, segmenting 5 activity clusters to isolate 209 bottleneck segments driving 35.8% processing overhead.
 - **Invariant Control:** Architected human-in-the-loop (HITL) gatekeepers and state invariant assertions over dynamic execution flows, validating 16 distinct operational branches with zero critical runtime failures.
- **Path-Finder: Heuristic & Algorithmic Traversal Engine** [GitHub](#) | [Live](#)
JavaScript, Data Structures, Graph Theory, Algorithm Optimization *2026*
 - **Optimal Graph Traversal:** Implemented search algorithms including A*, Dijkstra, BFS, and DFS utilizing custom min-heaps, securing theoretical optimal $O(V + E \log V)$ computation across 2,500+ node grids.
 - **Computational Profiling:** Decoupled heuristic search calculations from visual layout rendering via prioritized requestAnimationFrame scheduling queues, eliminating UI thread bottlenecking within 16ms frame intervals.
 - **Maze & Matrix Modeling:** Designed randomized recursive backtracker algorithms for non-trivial grid generation and state matrix export tools, reducing scenario synthesis latency by 70%.

LEADERSHIP EXPERIENCE

- **Sponsorship Head – Xpecto '26, IIT Mandi, Himachal Pradesh** *September 2025 – March 2026*
 - **Cross-Functional Operations:** Led a 25-member division managing stakeholder analytics, resource planning, and proposal pipelines, securing **INR 20+ Lakh** from 20+ national corporate and public sponsors.
- **Team Lead – Design Practicum, IIT Mandi, Himachal Pradesh** *August 2025 – December 2025*
 - **Applied Systems Engineering:** Led hardware-software integration for an IoT neonatal incubator prototype, driving end-to-end sensor data acquisition and lowering bill-of-materials to **INR 8,000** vs. the INR 1 Lakh commercial baseline.

HONORS & ACHIEVEMENTS

- **Hackathon Finalist: 1st Runner-Up** in Deep Learning Hackathon HACK60 for developing an end-to-end computer vision solution.
- **Competitive Programming:** Solved 500+ algorithmic problems across LeetCode and Codeforces (Rating: 1200+).
- **Inter-IIT Civil Conclave: 3rd Position** pan-IIT in the Sustainable City Planning Case Study at IIT Roorkee.